

Data Source Map

Audience: Engineering · Generated 2026-06-23

Where every dashboard panel gets its data, which database it lives in, and how the source/database badges on each panel are wired. Read this before touching anything that displays a number — the rule is **real data, an honest empty state, or a clearly-labelled mock; never a fabricated value dressed up as real.**

Last verified: 2026-06-21 (full scan of all 38 pages + 55 API routes).

1. The rule

Every data-bearing panel falls into one of four buckets. The first three are fine; the fourth is a bug.

Bucket	Meaning	Allowed?
Real	Pulled from a Supabase <code>_c</code> table, IRESS (via the worker), Yahoo, or an external API (SARB, ECB, RSS, Tavily).	Yes
Honest-empty	Renders <code>—</code> , <code>EmptyDataState</code> , "not configured", or an entitlement notice. No fake numbers.	Yes
Gated mock	Seed/demo data shown only behind a mock gate (<code>realDataOnly</code> , <code>IRESS_MODE=mock</code> , <code>USE_SUPABASE_QUOTES=false</code>) and labelled <code>MOCK / SEED</code> .	Dev only
Fabricated	Hardcoded / <code>Math.random</code> values rendered as if real, not gated, not labelled.	No — fix it

See section 7 for the current list of bucket-4 offenders.

2. Architecture / data flow

There is exactly **one IRESS CT licence seat**. The Vercel app never calls IRESS directly — a single Railway worker (`workers/iress-ingest`) holds the seat, polls IRESS, and writes the results into Supabase. The web app only ever reads Supabase (plus a few public external APIs).

IRESS V4 (CT seat)

```
└─ workers/iress-ingest (Railway, the only IRESS client)
  │ quotes loop      → securities_c.last_price, stock_intraday_c (retail)
  │                  │ quote_snapshot_c (institutional, full L1)
  │ └─ retail-ingest → securities_c.{last_price, change_percent, price_source} (DORMANT)
  └─ timeseries/bonds → yield_curve_history_c, oems_curve_metric_c, bonds_c,
                        index_intraday_c, sector_intraday_c (institutional)
```

Yahoo Finance

```
└─ /api/cron/yahoo-fundamentals → securities_c.{market_cap, pe_ratio, dividend_*, ytd_performance} (re
  (NEVER writes last_price/change_percent – IRESS owns those)
```

External (read live, no DB): SARB (rates), Frankfurter/ECB (FX), RSS (news), Tavily (AI research web)

Vercel app: page → /api/* (BFF) → Supabase _c tables / external API → render

Key gates (env):

- `IRESS_MODE` — `live` / `wsdl-stub` = real SOAP (worker only); anything else = mock. Vercel runs `mock` ; only the worker runs `live` .
- `USE_SUPABASE_QUOTES` — BFF reads quotes/curves/indices from Supabase vs in-process IRESS.
- `IRESS_RETAIL_INGEST=1` + `IRESS_RETAIL_DRY_RUN=0` — turns the full-universe retail price/day-move loop on (currently off → see §6).
- `NEXT_PUBLIC_USE_SUPABASE_QUOTES` — client real-data gate (`src/lib/data-policy.ts`). Defaults to **real-data-only** when unset; mock is an explicit opt-in (`=0` / `false` , or `?mock=1` with a DEV-MOCK banner). This is what keeps seed/fixture branches from rendering as if real — see §7.

3. The two databases

	RETAIL (mfxng...)	INSTITUTIONAL (nnwz...)
Client	<code>createRetailServiceRoleClient()</code>	<code>createInstitutionalServiceRoleClient()</code> / <code>createServiceRoleClient()</code>
What	LIVE retail platform + the shared price tables	OEMS desk trading book + desk analytics
Price tables	<code>securities_c</code> (universe + Yahoo fundamentals + IRESS price overlay), <code>stock_intraday_c</code>	<code>quote_snapshot_c</code> (IRESS full L1), <code>index_intraday_c</code> , <code>sector_intraday_c</code>
Curves/FI	—	<code>yield_curve_history_c</code> , <code>oems_curve_metric_c</code> , <code>bonds_c</code>
Books	<code>stock_holdings_c</code> , <code>strategies_c</code> , <code>client_strategy_returns_c</code> , <code>profiles</code> , KYC, <code>wallets</code>	<code>oems_order_audit</code> , <code>oems_account_c</code> , <code>oems_position_c</code> , <code>oems_transaction_c</code>
Other	<code>News_articles</code> (Alliance wire), <code>email_*</code> , <code>app_settings</code>	<code>macro_*</code> , <code>money_market_*</code> , <code>jibar_fixing_c</code> , <code>news_item_c</code> , <code>integration_worker_health</code>

`securities_c` and `stock_intraday_c` exist in **both** DBs with different shapes. The app's price reads hit the **retail** copy; the IRESS overlay is read from the **institutional** `quote_snapshot_c` and merged by `symbol` (the row ids differ per project). Legacy single `SUPABASE_URL` / `SUPABASE_SERVICE_ROLE_KEY` resolves to **institutional**.

4. The source badge

Defined in `src/components/oems/primitives/data-source-badge.tsx`, rendered by `GlassSection` (`glass.tsx`). A panel sets two props:

- `db` — "retail" (chip `MFNG`) or "institutional" (chip `NNWZ`). Omit for external feeds (SARB/ECB) that aren't in either DB.
- `dataSource` — a `DataSourceKind`: `iress` | `yahoo` | `external` | `supabase` | `live` | `seed` | `mock` | `hybrid` | `worker` | `stream` | `unconfigured` | `unavailable` | `blocked-external` | `blocked-vendor` | `code-gap`.

For panels whose source varies at runtime (the equities universe can be IRESS or Yahoo per row), pass `dataSource={mapSource(data?.source, "<default>")}` — `mapSource` (`src/lib/data-source.ts`) normalises a BFF's `source` / `price_source` string into a kind. Static panels (curves are always IRESS, admin is always Supabase) pass a literal kind. Entitlement/empty branches use `blocked-external` / `blocked-vendor`, not the populated-state kind.

Everything reads `SEED` / `MOCK` on a local dev server (`IRESS_MODE=mock`, no worker); the same panels resolve to `IRESS` / `YAHOO` / `SUPABASE` / `EXTERNAL` in production where the worker + crons are running.

5. Per-module map

`db` · `source` per page. "(real)" = the production branch; mock branches are gated + labelled.

OEMS desk

Page	Panel	Endpoint	DB	Source
Cockpit	AUM / Day P&L / Portfolio accounts	<code>/api/client-book</code>	retail	supabase
Cockpit	Open orders, blotter	<code>/api/orders</code>	institutional	supabase (IRESS order mirror)
Cockpit	Rates / FX tiles	<code>/api/sa-rates</code> , <code>/api/fx/*</code>	—	external (SARB, ECB)
Cockpit	Sector treemap, top movers, JSE heatmap	<code>/api/equities</code>	retail (+inst overlay)	iress / yahoo (runtime)
Cockpit	US heatmap / movers	<code>/api/global-movers</code>	—	yahoo
Cockpit	ZAR curve, curve-move PCA	<code>/api/curves/ZAR_NSS(/metrics)</code>	institutional	iress
Cockpit	J203 ALSI intraday	<code>/api/indices/J203</code>	institutional	blocked-external (TS2)
Cockpit	News flow	<code>/api/news</code>	retail	external (RSS + wire)
Cockpit	IPS accounts / positions	<code>/api/portfolio</code>	institutional	supabase
Equities	KPIs	<code>/api/portfolio</code>	institutional	supabase
Equities	Top movers, securities universe	<code>/api/equities</code>	retail (+inst)	iress / yahoo (runtime); 1M/6M cols TS2-blocked
Curves	all curve/PCA/OIS/carry panels	<code>/api/curves/*</code>	institutional	iress (seed/blocked when empty)
Fixed Income	bond screener, detail, P&L sensitivity	<code>/api/bonds</code>	institutional	iress (blocked-vendor when empty)

Page	Panel	Endpoint	DB	Source
Fixed Income	ZAR govt history	/api/curves/ZAR_GOVI	institutional	blocked-external (TS2)
Money Market	SARB rates	/api/sa-rates	—	external
Money Market	JIBAR, instruments	/api/money-market	institutional	blocked-external (TS2)
Macro	indicators, releases	/api/macro , /api/sa-rates	institutional / —	supabase / external
News	tape (All/SENS/Wires)	/api/news	retail	external (RSS + wire); SENS = blocked-vendor
Security	key stats	/api/quote-snapshot/[sym] + /api/equities	institutional + retail	iress + yahoo
Security	intraday / history chart	/api/intraday/[sym] , /api/history/[sym]	retail / worker	iress; history TS2-blocked
Security	depth L2 / time & sales	—	—	unconfigured (gated mock locally)
Blotter	orders + state	/api/orders , /api/orders/live	institutional	supabase / iress
Integration	worker health, latency, IRESS status	/api/worker-health	institutional	worker heartbeat
Research Lab	composition / fundamentals	/api/research-lab	retail	supabase
Research Lab	AI Research	/api/research-ai	retail (+inst cache)	yahoo + tavily + MiniMax-M3

Admin (all retail · supabase)

Dashboard, Clients, Order-book, Factsheets, Studio, Investors, EFT/Wallets, Emailers, Team, App-settings, Cyber-compliance, Mint-mornings, Approvals — 34 routes, all read/write the **retail** DB. No IRESS/Yahoo. Real where the tables are populated; honest-empty otherwise.

6. IRESS vs Yahoo — capability & gaps

IRESS is the source of truth. Yahoo is a fallback only where IRESS can't supply yet. Curves and bonds were confirmed live on **Exchange=YFX, DataSource=YFXD** (2026-06-16) — the older `DB_TOPOLOGY_DECISION.md` calling them blocked is stale.

Feed	IRESS now	Fallback	To go IRESS-only
Live price, intraday, full L1, day-move	yes	Yahoo (uncovered syms)	flip retail-ingest (no entitlement)
ZAR curves, bonds	yes (YFX/YFXD)	seed	done
Orders (live + audit)	yes (worker)	—	—
Fundamentals (mkt cap, P/E, div)	no	Yahoo cron	Charles: <code>SecurityGet</code> entitlement

Feed	IRESS now	Fallback	To go IRESS-only
YTD / returns history	no	Yahoo	Charles: <code>TimeSeriesGet2</code> equity history
Index intraday (J203), sectors, money market	no (TS2 on JSE/JSED)	seed	Charles: TS2 + sector/index codes
News / SENS	no	RSS + Alliance wire	SENS vendor (or confirm IRESS path)
L2 depth / time & sales	no	—	<code>PricingQuoteExGet</code> entitlement / vendor

Day-move is flat in production because the full-universe `retail-ingest` loop (which computes `change_percent` from `prev_close`) is built but dormant. Turning it on needs no entitlement: apply the `price_source` migration → set `IRESS_RETAIL_INGEST=1` + `RETAIL_SUPABASE_URL` → shadow run → flip `IRESS_RETAIL_DRY_RUN=0` → redeploy the worker.

7. Fabricated data — resolved (2026-06-21)

A full scan (all 38 pages + 55 routes) found the main desk + admin + BFFs clean. Seven fabrications lived in the secondary persona portals + settings; all are now fixed. The shared root cause was `isRealDataOnlyClient()` defaulting to NOT-real when `NEXT_PUBLIC_USE_SUPABASE_QUOTES` was unset, so the demo/fixture branches rendered under `SUPABASE / IRESS` badges.

#	File	What it fabricated	Fix applied
1	<code>src/app/settings/page.tsx</code>	IRESS endpoint health (<code>Math.random</code> p95/status) — ungated	query gated <code>enabled: !realDataOnly</code> ; real mode shows an honest note pointing to live worker health on the Integration page
2	<code>src/app/business/page.tsx</code>	named compliance breaches + sales pipeline	<code>PersonaRealDataGate</code> → honest "Not configured" by default (gate flip)
3	<code>src/app/compliance/page.tsx</code>	pending-approvals queue + audit trail	same — honest-empty by default
4	<code>src/app/fc/overview/page.tsx</code>	reconciliation legs / cash / exceptions	same — honest-empty by default
5	<code>src/app/wm/page.tsx</code>	client book (named clients, R250k withdrawal)	same — honest-empty by default
6	<code>src/app/strategist/page.tsx</code>	seed fixtures badged real	same — honest-empty by default
7	<code>src/app/admin/factsheets/page.tsx</code>	<code>cashWeight = 8</code> hardcoded	reads <code>app_settings.executionReserveRate</code> (defensive, falls back to 8)

The systemic fix (`src/lib/data-policy.ts`): `isRealDataOnlyClient()` now defaults to **true** (real-data-only) when the flag is unset, server- and client-side identically. Mock/seed is an explicit opt-in (`=0` / `false` or `?mock=1`). This makes all five persona pages render `PersonaRealDataGate` 's honest "Not configured" state by default, and flips the whole app to real data / honest-empty unless mock is deliberately requested. Verified live: `/wm` shows "Not configured" (no fake

clients); `/oems` cockpit renders real MFXNG·SUPABASE / NNWZ·IRESS / EXTERNAL / honest BLOCKED-EXTERNAL / UNCONFIGURED ; `/settings` shows the honest worker-health note.

Remaining minor hygiene (gated mock, not fabrication): `depth-ladder.tsx` + the time-&-sales panel use `Math.random` for a simulated order book — now only reachable under an explicit `?mock=1` opt-in (the real branch is honest-empty). An inline `SIMULATED` tag would be nice-to-have.

8. Page status — verified / pending / unverified (2026-06-21)

Checked page-by-page against the running dev server in **real-data-default mode with no IRESS worker** — so worker-fed feeds are empty/seed locally and populate in production. Method: every BFF endpoint probed for HTTP status / source / row count, and every page client-rendered and checked for a crash.

Nothing is broken — every endpoint returns 200, every page renders without a React error. The gaps below are "real path, no data here yet" (worker / entitlement / admin auth), not failures.

Legend: ✓ real data · ● honest-empty, awaiting worker/entitlement/data · △ unverified · X broken (none found)

OEMS desk — all render, no crashes

Page	Endpoint(s)	Local data	Notes
Cockpit <code>/oems</code>	client-book, orders, equities, curves, indices, news, portfolio, sa-rates, fx	✓ equities/orders(18)/news/rates · ● curves & J203 = seed · ● portfolio empty	curves are IRESS in prod, seed-fallback locally
Equities <code>/oems/equities</code>	<code>/api/equities</code> , <code>/api/portfolio</code>	✓ 246 securities · ● portfolio empty	1M/6M return cols pending TS2
Curves <code>/oems/curves</code>	<code>/api/curves/*</code> , <code>/metrics</code>	● seed (12 pts); metrics <code>unavailable</code>	IRESS-live in prod (YFX/YFXD)
Fixed Income <code>/oems/fixed-income</code>	<code>/api/bonds</code> , <code>/api/curves/ZAR_GOVI</code>	● bonds empty (0); govi history TS2-blocked	IRESS-live in prod
Money Market <code>/oems/money-market</code>	<code>/api/money-market</code> , <code>/api/sa-rates</code>	✓ SARB rates · ● instruments/JIBAR empty	TS2-blocked
Macro <code>/oems/macro</code>	<code>/api/macro</code> , <code>/api/sa-rates</code>	✓ SARB · ● indicators empty	vendor table empty locally
News <code>/oems/news</code>	<code>/api/news</code>	✓ RSS wire (5+)	SENS = vendor, not wired
Blotter <code>/oems/blotter</code>	<code>/api/orders (/live)</code>	✓ 18 orders	live-orders = worker passthrough
Security <code>/oems/security</code>	quote-snapshot, intraday, history, equities	✓ snapshot · ● intraday/history empty (no worker)	depth/T&S simulated only under <code>?mock=1</code>
Integration <code>/oems/integration</code>	<code>/api/worker-health</code>	● no worker locally	live in prod
Research Lab <code>/oems/research-lab</code>	<code>/api/research-lab</code> , <code>/api/research-ai</code>	✓ 9 strategies; AI works (MiniMax-M3)	

Portals & settings — all render

Page	Endpoint(s)	Local data	Notes
Strategies /strategies	/api/strategies	✓ 9 strategies (real)	the real StrategiesMonitor
WM / Strategist / FC / Business / Compliance	PersonaRealDataGate	🕒 honest "Not configured" by default	⚠ headers show demo identities — see Unsure #4
Settings /settings	iress.config + worker-health note	✓ config real; health = honest note	

Admin — render, but data UNVERIFIED

All /admin/* pages (dashboard, clients, order-book, factsheets, emailers, team, app-settings, investors, eft) render without crashing, but under the session used here they show the **auth gate** — the admin BFFs require admin RBAC and returned `no-session` (401) both headless and in-browser. ⚠ **Their data rendering needs an authenticated admin session to verify — not checked.**

Unsure / needs attention

1. **Admin page data** — unverified; needs an admin login (the RBAC gate blocked verification here).
2. **Worker-fed feeds** — live curves, intraday, history, portfolio/IPS accounts, money-market, J203 index, sectors are empty/seed locally because the Railway `iress-ingest` worker isn't running. **Unverified against a live worker**; expected to populate in production once the worker + entitlements are on.
3. **Curves / indices / sectors badges** — show `IRESS / blocked-external` (the *production* source), but locally without the worker the data is a **seed fallback**. The badge is prod-accurate, not local-accurate.
4. **Persona headers** — `wm/strategist/fc/business/compliance` data is honest-empty, but `PersonaHeader` still shows a **sample demo identity** from `PERSONA_USERS` (`src/lib/store/session-provider.ts`) — the persona-switcher demo. Product call: keep the demo personas or wire real auth identities. (Not financial data.)
5. `/api/history` — returns "Railway IRESS worker URL not configured" locally → honest-empty; needs `IRESS_WORKER_URL` set in production.
6. **SENS feed** — not configured; needs a vendor (or a confirmed IRESS news path).

Not working

None found. Every endpoint returns 200; every page renders without a crash. The empties are honest, expected states — not bugs.

9. Source-of-truth files

- Supabase clients + DB targeting: `src/lib/supabase/server.ts`
- Source kinds + badge: `src/components/oems/primitives/data-source-badge.tsx`
- Badge mapping helper: `src/lib/data-source.ts`
- Real-data gate (client): `src/lib/data-policy.ts`
- Quote routing / `deriveDataSource` : `src/lib/hooks/quote-routing.ts`
- IRESS adapter + mock: `src/lib/iress/`
- IRESS worker: `workers/iress-ingest/`
- Yahoo fundamentals cron: `src/app/api/cron/yahoo-fundamentals/route.ts`
- AI research (Yahoo + Tavily + MiniMax): `src/lib/research-ai/`

